

at the air-dielectric separation surface. The waves trapped in the substrate are generally undesirable.

Different types of antennae have many different shapes and dimensions. Microstrip antennae can be classified into four subtypes: microstrip patch antennae, microstrip dipoles, printed slot antennae and microstrip travelling-wave antennae.

Polarisation and radiation of microstrip antennae

The polarisation of an antenna is determined by the wave radiated in a given direction that is identical to the direction of the electric field. The instantaneous electric field vector traces a figure in time. This figure is usually an ellipse that has special cases. If the path of the electric field vector follows a line, the antenna is said to be linearly polarised. If the electric field vector rotates in a circle, it is called circularly polarised. To characterise the polarisation, axial ratio is used. It is defined by

TABLE II
Comparison of Various Types of Microstrip Antennae

Characteristics	Microstrip patch antenna	Microstrip slot/travelling-wave antenna	Printed dipole antenna
Profile	Thin	Thin	Thin
Fabrication	Very easy	Easy	Easy
Polarisation	Both linear and circular	Both linear and circular	Linear
Dual-frequency operation	Possible	Possible	Possible
Shape flexibility	Any shape	Mostly rectangular and circular shape	Rectangular and triangular
Spurious radiation	Exists	Exists	Exists
Bandwidth	2-50 per cent (resonant frequency)	5-30 per cent (resonant frequency)	30 per cent (resonant frequency)

the following relationship:

$$T = \frac{\text{Large diameter of the ellipse}}{\text{Small diameter of the ellipse}}$$

Polarisation is said to be linear if $T \rightarrow \infty$ or $T = 0$, and circular if $T = 1$.

There are two sub-types of linear polarisation: Vertical polarisation and horizontal polarisation. When 'E' field vector of the EM wave is perpendicular

to the earth, the EM wave is said to be vertically polarised. When 'E' field vector of the EM wave is parallel to the earth, the EM wave is said to be horizontally polarised.

The radiation pattern is a graphical depiction of the relative field strength transmitted from or received by the antenna. Antenna radiation patterns



PSI Enterprises
314, Krishna Mall, Sector-12
Dwarka, New Delhi - 110075
Tel: +91-9121325888
www.psindia.com
sources2004@yahoo.co.in

Fax: +91-45649630
Email: sales@psindia.com

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Peenya 2nd Stage, Bangalore - 560 056, (India)
Tel: +91 80 28362750/ 51 (Fax: +91 80 2836 2752
sales@miracle.net.in | www.toroidal.com